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ITA0513 – COMPUTER VISION

Name: A. Midhuna

Reg no: 192521302

ASSIGNMENT – 2

Image Enhancement and Segmentation

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA0513	Computer Vision	Assignment – 2	CO2	BTL3	50

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Bit-plane and intensity-level slicing	3	3	2	2	10
2	Histogram specification	3	3	2	2	10
3	Gaussian smoothing and unsharp masking	3	3	2	2	10
4	Discrete Fourier transform filtering	3	3	2	2	10
5	Otsu optimal thresholding	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO2, BTL3, 10 Marks) – Bit-Plane and Intensity-Level Slicing for Cheque Forgery Analysis

A bank's forensic unit examines a scanned 8-bit cheque suspected of alteration. The forger is believed to have overwritten the amount with ink whose scanned intensity lies between 140 and 190, while the original ink is darker. The bank also embeds an invisible security watermark in one bit plane. Six pixels sampled across the amount field are listed below. The analyst must decide how to store a compact copy of the evidence and how to highlight the suspicious ink.

Pixel	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**
Intensity r	214	181	150	92	37	165

(a) Express each pixel in 8-bit binary and determine the value of bit plane 7 (MSB) and bit plane 0 (LSB) for every pixel. [3 Marks]

(b) Reconstruct each pixel using only bit planes 7 to 4 (a) by setting the lower nibble to 0 and (b) by setting the lower nibble to its midpoint 8. Compute the mean squared error (MSE) of both reconstructions. [3 Marks]

(c) Apply intensity-level slicing to highlight the suspicious range [140, 190] (i) by setting it to 255 and preserving other levels, and (ii) by producing a binary map (255 inside, 0 outside). Identify the suspicious pixels. [2 Marks]

(d) The negative transformation $s = 255 - r$ is also proposed. Compute it for P4 and P5, and justify in which bit plane the watermark is most likely hidden and how that plane should be examined. [2 Marks]

ANSWER:

Bit-Plane and Intensity-Level Slicing for Cheque Forgery Analysis

(a) 8-bit binary representation and bit planes

Each pixel intensity is converted into an 8-bit binary number.

Pixel	Intensity (r)	8-bit binary	Bit 7 (MSB)	Bit 0 (LSB)
P1	214	11010110	1	0
P2	181	10110101	1	1
P3	150	10010110	1	0
P4	92	01011100	0	0
P5	37	00100101	0	1
P6	165	10100101	1	1

Bit plane 7 (MSB): Represents the highest intensity contribution (128).

Bit plane 0 (LSB): Represents the smallest intensity contribution (1).

- Bit plane 7 values: 1, 1, 1, 0, 0, 1.
- Bit plane 0 values: 0, 1, 0, 0, 1, 1.

(b) Reconstruction using bit planes 7 to 4

Only the upper four bits are retained.

(i) Lower nibble set to 0

Formula:

$$R_1=(r>>4)\times16$$

(ii) Lower nibble set to midpoint 8

Formula:

$$R_2=(r>>4)\times16+8$$

Reconstruction table:

Pixel	Original	Lower nibble = 0	Lower nibble = 8
P1	214	208	216
P2	181	176	184
P3	150	144	152
P4	92	80	88
P5	37	32	40
P6	165	160	168

Mean Squared Error (MSE)

Formula:

$$MSE=\frac{\sum(r-R)^2}{N}$$

For (i), lower nibble = 0:

$$MSE=\frac{291}{6}$$

$$\boxed{MSE=48.5}$$

For (ii), lower nibble = 8:

$$MSE=\frac{51}{6}$$

$$\boxed{MSE=8.5}$$

Therefore, reconstruction using midpoint 8 gives lower error and better image quality.

(c) Intensity-level slicing

Given suspicious intensity range:

$$140\leq r\leq190$$

(i) Highlight suspicious pixels by setting them to 255 and preserving others.

Formula:

\[s= \begin{cases} 255 & 140\leq r\leq190\\ r & \text{otherwise} \end{cases} \]

Pixel	Original	Output
P1	214	214
P2	181	255
P3	150	255
P4	92	92
P5	37	37
P6	165	255

(ii) Binary map

Formula:

\[s= \begin{cases} 255 & 140\leq r\leq190\\ 0 & \text{otherwise} \end{cases} \]

Pixel	Binary output
P1	0
P2	255
P3	255
P4	0
P5	0
P6	255

Suspicious pixels: P2, P3 and P6

These pixels may indicate overwritten ink.

(d) Negative transformation and watermark

Given formula:

\[s=255-r \]

For P4:

\[s=255-92=\boxed{163} \]

For P5:

\[s=255-37=\boxed{218} \]

Watermark identification:

- The watermark is most likely hidden in bit plane 0 (LSB).
- LSB changes produce very small intensity differences, making the watermark difficult to notice visually.
- The analyst should extract and examine each bit plane separately, especially bit plane 0, to identify the hidden watermark pattern.
- The original evidence should be preserved without modification.

- **Conclusion:** Bit-plane slicing helps identify hidden watermark information, while intensity-level slicing highlights suspicious ink. These techniques help detect possible cheque forgery

Question 2 (CO2, BTL3, 10 Marks) – Histogram Specification for Consistent Endoscopy Images

A hospital standardises the appearance of 3-bit (L = 8) endoscopy thumbnails of size 32 × 32 so that images from different devices look alike to a diagnostic classifier. A thumbnail from an older device has the input histogram below, and the reference histogram $p_z(z)$ recommended by the clinicians is also given. Use $s_k = \text{round}[7 \cdot \Sigma p_r]$, $G(z_q) = \text{round}[7 \cdot \Sigma p_z]$, and map each s_k to the smallest z_q whose $G(z_q)$ is closest to s_k.

Level	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Input count n_k	180	260	220	150	100	60	34	20
Target p_z	0	0	0.05	0.10	0.20	0.30	0.20	0.15

(a) Compute the input probabilities, their cumulative sum and the equalised values s_k. [3 Marks]

(b) Compute the transformation $G(z_q)$ for the target histogram. [3 Marks]

(c) Determine the mapping $r_k \rightarrow z_q$ and the resulting output histogram. [2 Marks]

(d) Compare the achieved histogram with the target and justify why specification, rather than equalisation, is appropriate for this application. [2 Marks]

ANSWER:

Histogram Specification for Consistent Endoscopy Images

Given:

- Image size = \(32\times32=1024\) pixels
- Number of intensity levels = \(L=8\)
- Input histogram counts = 180, 260, 220, 150, 100, 60, 34, 20.

(a) Input probabilities, cumulative sum and equalised values

Step 1: Calculate input probabilities

Formula:

\[p_r(r_k)=\frac{n_k}{1024} \]

Step 2: Calculate cumulative sum

\[C(r_k)=\sum_{j=0}^{k}p_r(r_j) \]

Step 3: Calculate equalised values

\[s_k=round[7\times C(r_k)] \]

Level	Count	Probability	Cumulative sum	Equalised value
0	180	0.1758	0.1758	1
1	260	0.2539	0.4297	3
2	220	0.2148	0.6445	5
3	150	0.1465	0.7910	6
4	100	0.0977	0.8887	6
5	60	0.0586	0.9473	7
6	34	0.0332	0.9805	7
7	20	0.0195	1.0000	7

(b) Target transformation G(zq)

Given formula:

\[G(z_q)=round[7\times\sum p_z(z_q)] \]

Calculate the cumulative probability of the target histogram.

Target level	**Target probability**	**Cumulative probability**	**G(zq)**
0	**0.00**	**0.00**	**0**
1	**0.00**	**0.00**	**0**
2	**0.05**	**0.05**	**0**
3	**0.10**	**0.15**	**1**
4	**0.20**	**0.35**	**2**
5	**0.30**	**0.65**	**5**
6	**0.20**	**0.85**	**6**
7	**0.15**	**1.00**	**7**

(c) Determine mapping and resulting output histogram

Mapping rule:

Map each input equalised value \(s_k\) to the smallest target level whose \(G(z_q)\) is closest to \(s_k\).

Inputlevel (rk)	Equalised value (sk)	Mapped target level (zq)
0	1	3
1	3	4
2	5	5
3	6	6

4	6	6
5	7	7
6	7	7
7	7	7

Resulting output histogram:

Combine the input counts that map to the same target level.

- Level 3 = 180
- Level 4 = 260
- Level 5 = 220
- Level 6 = 150 + 100 = 250
- Level 7 = 60 + 34 + 20 = 114

Output level	Output count
0	0
1	0
2	0
3	180
4	260
5	220
6	250
7	114
Total	1024

(d) Comparison and justification

Comparison:

- The achieved histogram does not exactly match the target histogram.
- This happens because of rounding and the limited number of intensity levels (8 levels).
- Multiple input levels are mapped to the same output level.

Why histogram specification is preferred:

- Histogram specification matches the image appearance to a predefined reference histogram.
- It maintains consistent brightness and contrast across different endoscopy devices.
- The reference histogram is recommended by clinicians, making it suitable for medical image standardisation.
- Histogram equalisation only improves the overall contrast and does not follow a specific target appearance.

Conclusion: Histogram specification is more appropriate because it produces images with a desired intensity distribution, helping maintain consistency for diagnostic classification.

Question 3 (CO2, BTL3, 10 Marks) – Gaussian Smoothing and Unsharp Masking for Weld Radiographs

A pipeline contractor inspects weld radiographs for lack-of-fusion defects that appear as faint dark lines beside bright weld beads. The image contains Gaussian sensor noise, so the engineer proposes a 3 × 3 Gaussian kernel with $\sigma = 1$, $w(x, y) = \exp[-(x^2 + y^2)/(2\sigma^2)]$, normalised to unit sum, followed by unsharp masking $g = f + k(f - \bar{f})$. The 3 × 3 neighbourhood centred on a pixel at the defect boundary is shown below.

Row / Col	**C1**	**C2**	**C3**
R1	88	92	140
R2	85	96	150
R3	90	94	146

(a) Compute the unnormalised and normalised Gaussian kernel weights (centre, edge and corner) and verify the sum. [3 Marks]

(b) Apply the Gaussian kernel and a 3 × 3 box filter at the centre pixel and compare the results. [3 Marks]

(c) Apply unsharp masking with k = 1 and high-boost filtering with k = 2.5 at the centre pixel, using the Gaussian result as $\bar{f}$. [2 Marks]

(d) The noise-variance reduction factor of a linear filter is Σw^2. Compute it for the Gaussian and box kernels, and justify the filter and k value for defect detection. [2 Marks]

ANSWER:

Gaussian Smoothing and Unsharp Masking for Weld Radiographs

Given:

- Image neighbourhood = 3 × 3
- Gaussian standard deviation, $\sigma=1$
- Unsharp masking $k=1$
- High-boost filtering $k=2.5$

Input image:

$$\begin{bmatrix} 88&92&140\\ 85&96&150\\ 90&94&146 \end{bmatrix}$$

(a) Unnormalised and normalised Gaussian kernel weights

Given formula:

\[w(x,y)=e^{-\frac{x^2+y^2}{2\sigma^2}} \]

Since \(\sigma=1\):

1. Calculate weights

- Centre (0,0):

\[e^0=1 \]

- Edge (1,0):

\[e^{-1/2}=0.6065 \]

- Corner (1,1):

\[e^{-1}=0.3679 \]

Therefore, the unnormalised Gaussian kernel is:

\[\begin{bmatrix} 0.3679&0.6065&0.3679\\ 0.6065&1&0.6065\\ 0.3679&0.6065&0.3679 \end{bmatrix} \]

2. Calculate total sum

\[S=1+4(0.6065)+4(0.3679) \]

\[S=4.8976 \]

Normalised Gaussian kernel:

Divide each weight by 4.8976.

- Centre = \(1/4.8976=0.2042\)
- Edge = \(0.6065/4.8976=0.1238\)
- Corner = \(0.3679/4.8976=0.0751\)
- Normalised Gaussian kernel:
- \[\boxed{ \begin{bmatrix} 0.0751&0.1238&0.0751\\ 0.1238&0.2042&0.1238\\ 0.0751&0.1238&0.0751 \end{bmatrix} } \]

(b) Gaussian filter and 3 × 3 box filter

1. Gaussian filtering

Formula:

\[g(x,y)=\sum w(x,y)f(x,y) \]

Substitute the pixel values and normalised weights:

$$\begin{aligned} g = {} & (88 \times 0.0751) + (92 \times 0.1238) \\ & +(140 \times 0.0751) + (85 \times 0.1238) \\ & +(96 \times 0.2042) + (150 \times 0.1238) \\ & +(90 \times 0.0751) + (94 \times 0.1238) \\ & +(146 \times 0.0751) \end{aligned}$$

$$\boxed{Gaussian\ result = 106.59}$$

2. Box filtering

A 3 × 3 box filter gives equal weight to all nine pixels.

$$ Box\ filter=\frac{\text{Sum of all pixels}}{9} $$

$$ =\frac{88+92+140+85+96+150+90+94+146}{9} $$

$$ \boxed{Box\ result=109} $$

Comparison:

- Gaussian filter gives more importance to the centre pixel.
- Box filter gives equal importance to all pixels.
- Gaussian filtering provides smoother results while preserving local structure better than a box filter.

(c) Unsharp masking and high-boost filtering

Given:

Original centre pixel:

$$ f=96 $$

Gaussian result:

$$ \bar f=106.59 $$

Formula:

$$ g=f+k(f-\bar f) $$

1. Unsharp masking (k = 1)

$$ g=96+1(96-106.59) $$

$$ \boxed{g=85.41} $$

2. High-boost filtering (k = 2.5)

$$ g=96+2.5(96-106.59) $$

$$ \boxed{g=69.52} $$

Therefore, high-boost filtering produces stronger sharpening than unsharp masking.

(d) Noise variance reduction and filter justification

Formula:

$$ Variance\ reduction\ factor=\sum w^2 $$

1. Gaussian filter

$$ =4(0.0751)^2+4(0.1238)^2+(0.2042)^2 $$

$$ \boxed{0.1256} $$

2. Box filter

Each weight = 1/9.

$$ =9\times(1/9)^2 $$

$$ \boxed{0.1111} $$

Justification:

- Gaussian filtering reduces Gaussian sensor noise and preserves image structure better than a box filter.
- Box filtering provides slightly greater noise reduction but can blur important defect boundaries.
- Unsharp masking with k = 1 provides moderate sharpening.
- High-boost filtering with k = 2.5 may amplify noise and create unwanted edges.

Conclusion: Gaussian smoothing followed by unsharp masking with k = 1 is a suitable choice for detecting faint defects while maintaining image details.

Question 4 (CO2, BTL3, 10 Marks) – DFT-Based Filtering of a Textile Line-Scan Signal

A textile mill uses a line-scan camera to monitor fabric uniformity. To validate a frequency-domain filter on the embedded processor, the engineer analyses a short four-sample scan line f(x) = [52, 108, 140, 92] for x = 0 to 3. The DFT is $F(u) = \Sigma f(x) \cdot e^{-j2\pi ux/4}$ and the inverse is $f(x) = (1/4)\Sigma F(u) \cdot e^{j2\pi ux/4}$. The highest frequency component (u = 2) represents the alternating weave pattern, while u = 1 and u = 3 represent slow shading.

(a) Compute F(u) for u = 0, 1, 2 and 3. [3 Marks]

(b) Compute the power spectrum $|F(u)|^2$ and the percentage of total energy in each component. Interpret the result. [3 Marks]

(c) Apply an ideal low-pass filter that removes u = 2 and compute the filtered line. Then apply a filter that removes only the DC term and compute the result. [2 Marks]

(d) Using the results, justify which component the uniformity monitor should threshold to detect a weave defect, and state why a practical 2-D implementation must pad the image before filtering. [2 Marks]

ANSWER:

DFT-Based Filtering of a Textile Line-Scan Signal

Given:

$$ f(x)=[52,108,140,92] $$

Number of samples:

$$ N=4 $$

DFT formula:

$$ F(u)=\sum_{x=0}^{3}f(x)e^{-j2\pi ux/4} $$

(a) Compute DFT values

1. For u = 0

$$ F(0)=52+108+140+92 $$

$$\boxed{F(0)=392}$$

2. For u = 1

$$F(1)=52+108(-j)+140(-1)+92(j)$$

$$=52-108j-140+92j$$

$$\boxed{F(1)=-88-16j}$$

3. For u = 2

$$F(2)=52-108+140-92$$

$$\boxed{F(2)=-8}$$

4. For u = 3

$$\boxed{F(3)=-88+16j}$$

Final DFT values:

u	DFT value F(u)
0	392
1	-88 - 16j
2	-8
3	-88 + 16j

(b) Power spectrum and percentage of energy

Formula:

$$|F(u)|^2=Re^2+Im^2$$

1. Calculate power spectrum

For u = 0:

$$|F(0)|^2=392^2=153664$$

For u = 1:

$$|F(1)|^2=(-88)^2+(-16)^2$$

$$=7744+256=8000$$

For u = 2:

$$|F(2)|^2=(-8)^2=64$$

For u = 3:

$$|F(3)|^2=(-88)^2+16^2$$

$$=8000$$

2. Total energy

$$E=153664+8000+64+8000$$

$$\boxed{E=169728}$$

3. Percentage of energy

Formula:

\[Percentage=\frac{|F(u)|^2}{Total\ energy}\times100 \]

u	Power spectrum	Energy percentage
0	153664	90.535%
1	8000	4.713%
2	64	0.038%
3	8000	4.713%
Total	169728	100%

Interpretation:

- DC component (u = 0) contains the highest energy.
- Components u = 1 and u = 3 represent slow shading.
- Component u = 2 contains very little energy and represents the alternating weave pattern.

(c) Apply frequency-domain filters

Inverse DFT formula:

\[f(x)=\frac14\sum_{u=0}^{3}F(u)e^{j2\pi ux/4} \]

1. Ideal low-pass filter (remove u = 2)

Set:

\[F(2)=0 \]

Filtered DFT:

\[[392,\ -88-16j,\ 0,\ -88+16j] \]

Applying inverse DFT:

\[\boxed{f(x)=[54,106,142,90]} \]

2. Filter that removes only DC component

Set:

\[F(0)=0 \]

Filtered DFT:

\[[0,\ -88-16j,\ -8,\ -88+16j] \]

Applying inverse DFT:

\[\boxed{f(x)=[-46,10,42,-6]} \]

Comparison:

- Removing u = 2 removes the alternating weave pattern.
- Removing DC removes the average brightness value.
- The second filter produces a zero-mean signal.

(d) Component selection and 2-D padding

Component selection:

- The monitor should threshold the u = 2 component.
- It represents the alternating weave pattern.
- Its power spectrum can be monitored to identify changes caused by weave defects.
- If the power exceeds or falls below a suitable reference threshold, the system can flag a possible defect.

Why padding is required in 2-D filtering:

- Padding prevents opposite image boundaries from interacting due to the periodic nature of DFT.
- It reduces unwanted boundary effects and wraparound artifacts.
- It helps perform frequency-domain filtering more accurately.

Conclusion: DFT separates the signal into frequency components. By monitoring u = 2, the textile system can detect changes in the alternating weave pattern and identify possible fabric defects.

Question 5 (CO2, BTL3, 10 Marks) – Otsu Thresholding for Steel-Slab Scale Detection

A steel plant detects oxide scale on hot-rolled slabs. Scale appears darker than clean metal, and the image has been quantised to 3 bits (levels 0 to 7) for fast processing on a programmable logic controller. The histogram of a slab image with 680 pixels is given below. The quality engineer requires a threshold that maximises the between-class variance $\sigma_B^2(k) = [m_G P_1(k) - m(k)]^2 / \{P_1(k)[1 - P_1(k)]\}$.

Level	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Count	30	90	160	120	40	70	110	60

(a) Compute the normalised histogram, the global mean m_G and the global variance σ_G^2. [3 Marks]

(b) Compute $P_1(k)$, $m(k)$ and $\sigma_B^2(k)$ for k = 2, 3 and 4. [3 Marks]

(c) Given σ_B^2 = 0.556, 1.586, 2.769 and 1.206 for k = 0, 1, 5 and 6, determine the optimal threshold k* and the separability measure $\eta = \sigma_B^2(k^*)/\sigma_G^2$. [2 Marks]

(d) Interpret η and justify whether Otsu's method is suitable for the controller, noting how the result would change if smoothing were applied first. [2 Marks]

ANSWER:

Otsu Thresholding for Steel-Slab Scale Detection

Given:

- Total pixels = 680
- Intensity levels = 0 to 7

- Otsu threshold formula:

  \[\sigma_B^2(k)= \frac{[m_GP_1(k)-m(k)]^2} {P_1(k)[1-P_1(k)]} \]

  (a) Normalised histogram, global mean and variance

  1. Normalised histogram

  Formula:

$$p(r_k) = \frac{n_k}{680}$$

Level	Count	Probability
0	30	0.0441
1	90	0.1324
2	160	0.2353
3	120	0.1765
4	40	0.0588
5	70	0.1029
6	110	0.1618
7	60	0.0882
Total	680	1.0000

  2. Global mean

  Formula:

$$m_G = \frac{\sum r_k \, n_k}{N}$$

$$= \frac{0(30) + 1(90) + 2(160) + 3(120) + 4(40) + 5(70) + 6(110) + 7(60)}{680}$$

$$= \frac{2360}{680}$$

$$\boxed{m_G = 3.4706}$$

  3. Global variance

  Formula:

  \[\sigma_G^2=\frac{\sum n_k(r_k-m_G)^2}{N} \]

  \[\boxed{\sigma_G^2=4.2785} \]

  (b) Calculate P1(k), m(k) and between-class variance [3 Marks]

  Formulas:

  \[P_1(k)=\sum_{i=0}^{k}p(r_i) \]

  \[m(k)=\sum_{i=0}^{k}i\cdot p(r_i) \]

$$\sigma_B^2(k)= \frac{[m_GP_1(k)-m(k)]^2} {P_1(k)[1-P_1(k)]}$$

For k = 2

$$P_1(2)=\frac{30+90+160}{680} =0.4118$$

$$m(2)=\frac{0(30)+1(90)+2(160)}{680} =0.6029$$

$$\sigma_B^2(2)=2.8177$$

For k = 3

$$P_1(3)=\frac{30+90+160+120}{680} =0.5882$$

$$m(3)=\frac{0+90+320+360}{680} =1.1324$$

$$\sigma_B^2(3)=3.4126$$

For k = 4

$$P_1(4)=\frac{30+90+160+120+40}{680} =0.6471$$

$$m(4)=\frac{0+90+320+360+160}{680} =1.3676$$

$$\sigma_B^2(4)=3.3758$$

Result table:

k	P1(k)	m(k)	Between-class variance
2	0.4118	0.6029	2.8177
3	0.5882	1.1324	3.4126
4	0.6471	1.3676	3.3758

(c) Optimal threshold and separability measure

Given:

k	Between-class variance
0	0.556
1	1.586
2	2.818
3	3.413
4	3.376
5	2.769
6	1.206

The maximum between-class variance is 3.413 at k = 3.

Therefore,

$$\boxed{k^*=3}$$

Separability measure

Formula:

$$\eta=\frac{\sigma_B^2(k^*)}{\sigma_G^2}$$

$$=\frac{3.4126}{4.2785}$$

$$\boxed{\eta=0.7976}$$

(d) Interpretation and suitability

Interpretation:

- The separability measure is approximately 0.798 (79.8%).
- It indicates good separation between the dark scale and clean metal regions.
- Therefore, Otsu's method is suitable for detecting oxide scale.

Suitability for the controller:

- Otsu's method automatically selects the threshold.
- It requires simple histogram calculations.
- It is computationally efficient for a programmable logic controller (PLC).
- It does not require manually selecting the threshold.

Effect of smoothing:

- Smoothing reduces noise before thresholding.
- It can improve separation between scale and clean metal.
- The histogram and optimal threshold may change after smoothing.
- Excessive smoothing may remove small defects and blur boundaries.

Conclusion: Otsu's thresholding with optimal threshold k = 3 is suitable for detecting dark oxide scale efficiently.